

The influence of rainfall variability on the species composition of a northern California salt marsh plant assemblage

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Abstract

Natural variability in the species composition of salt marsh plant assemblage was studied for 4 years (1986–90) at Bolinas Lagoon, California (USA). The study was conducted during a period in which little or no physical disturbance occurred in the marsh. During the study, Bolinas Lagoon experienced highly variable rainfall, both within and between years. In years with average or below average winter and spring rainfall, the cover of *Salicornia virginica* increased relative to that of the other species. During the one year with much higher than average spring rainfall, the cover of *Salicornia* declined and rare species increased in relative abundance. These patterns suggest that under typical marsh conditions, the plant assemblage experiences stress due to low availability of freshwater and high soil salinity. Under such conditions, the stress-tolerant *Salicornia* dominates the assemblage. If, however, there is abundant rainfall early in the growing season, the other species are released from stress and are able to increase in cover. The marsh of Bolinas Lagoon appears to be resistant to change and able to tolerate stress. Few studies have examined natural variability in species in undisturbed marshes and more of such studies must be made in order to understand whether or not marshes are in general resistant to change caused by rainfall variability.

Abbreviations: MLLW = mean lower low water

Nomenclature: follows Munz, P.A. 1968. A California flora and supplement. Univ. California. Press. Berkeley.

Introduction

Concerns about predicting the impact of global climate change have lead ecologists to question how well we understand the dynamics of communities in the absence of such change (Mooney 1991). This is especially true for communities that

occur across a large geographic range. In the past, studies have commonly been conducted at a single site. However, results were often inferred to apply at all sites where similar communities occurred (Foster 1990). It is apparent that while communities which occur on different continents (Underwood and Fairweather 1986) and across large

geographic areas of one continent (Foster 1990) may exhibit similar patterns, the processes that generate the patterns may be very different. Our understanding of community dynamics is further complicated by small-scale variation within a geographic region. Information on such variation is often lacking (Foster 1990). Thus it is important to study natural year-to-year variation in communities at numerous sites to determine how general the observed patterns are.

Salt marshes occur along temperate and polar coastlines throughout the world (Chapman 1974). Because they are widespread and appear to have similar patterns throughout their range, they are an ideal system in which to study natural variability and examine the issue of generality. One agent of change in salt marsh plant assemblages is natural physical disturbance. Disturbance is any agent which causes complete or partial destruction of the vegetation canopy resulting in the creation of bare space (Grime 1977). The most commonly observed disturbances in salt marshes are sedimentation during storms, flooding, deposition of drift wood and mats of beach wrack, herbivory, and parasitism by dodder *Cuscuta salina* (Beeftink 1979; Niering & Warren 1980; Oliver & Reilly 1981; Hartman *et al.* 1983; Onuf & Quammen 1983; Zedler 1983; 1986; Zedler & Beare 1986; Zedler *et al.* 1980; 1986; Bertness & Ellison 1987; Ellison 1987; Onuf 1987).

This particular study was motivated by the question: Do salt marsh plant assemblages undergo changes in species composition over a short period of time (3 to 4 years) in the absence of disturbance? The initial expectation was that mature, upper marsh assemblages should not change much over this time scale (Redfield 1972). Most of the species which occur in California salt marshes are long lived perennials that reproduce and spread primarily by clonal growth (Munz 1968; Macdonald & Barbour 1974; Macdonald 1977). Long lived clonal plants are able to persist in a particular site for long periods of time once they are well established (Kershaw 1973; Lovett Doust 1981). On the other hand, it may be that species composition within each zone or at a particular site is not constant and that it can change

quickly even in the absence of disturbance. This could happen if the clones were spreading rapidly and growing around each other in a guerilla fashion (Lovett Doust 1981).

It should be noted that during this study, yearly rainfall was highly variable at Bolinas Lagoon. The California coast has historically experienced variable rainfall (Michealsen *et al.* 1987). The variable, low annual rainfall experienced by California from 1986 to the present (Climatological Data Annual Summaries 1986–90) have lead to drought conditions in the terrestrial environment. Lack of freshwater can place limits on the growth and reproduction of salt marsh plant species (Barbour & Davis 1970; Zedler & Beare 1986). Variation in the amount and timing of rainfall, both seasonally and during the tidal cycle, determines the magnitude of the effect of precipitation on salt marsh plants. The salt marsh is regularly inundated by water, albeit salt water, on a fairly regular basis. The stress is not so much due to a lack of water as to an accumulation of salt in marsh soils. Therefore a terrestrial drought does not automatically imply a drought in a salt marsh. Even so, variation in rainfall may lead to variation in the species composition of a salt marsh due to differential growth of the species (Zedler & Onuf 1984; Zedler & Beare 1986; Olff *et al.* 1988; De Leeuw *et al.* 1990).

Salt marshes are found along the Pacific coast of North America from Cabo San Lucas, Baja California, Mexico to the Aleutian Islands, U.S.A. (Macdonald & Barbour 1974). There have been numerous studies of these marshes but most of them have been short term (approximately 1 year) studies of zonation patterns, productivity, or growth conditions (summarized in Allison 1991). Longer studies (3 or more years) which have examined year-to-year changes in species composition and productivity have been concentrated in southern California (Purer 1942; Zedler 1982), especially in Tijuana Estuary (Zedler 1983; 1986; Zedler & Beare 1986; Zedler & Nordby 1986; Zedler *et al.* 1980; 1986) and Mugu Lagoon (Onuf 1987). Bolinas Lagoon is in northern California approximately 950 km northwest of Tijuana Estuary. Salt marshes in northern Cali-

ifornia differ from marshes in southern California in species composition and climatic conditions (Macdonald 1977). In general, northern California has more rainfall and lower temperatures than southern California. This is a unique multi-year study of natural variability of a northern California salt marsh and adds to our understanding of geographical variability among North American Pacific coast salt marshes.

Methods

Study site

This research was conducted in the salt marsh at Bolinas Lagoon, a small coastal estuary 24 km northwest of San Francisco, California. The lagoon experiences daily tidal inundation. The research sites were located on Kent Island in the center of the lagoon (see map in Allison 1991). Kent Island is a low-lying island of approximately 40 ha. covered with salt marsh vegetation, except for a small area of dunes inhabited by dune vegetation on the western margin of the island. The island was chosen as the main study site because it has extensive areas of marsh, is relatively isolated and thus is seldom disturbed by human activity. The recent geologic history of Bolinas Lagoon is summarized in Rowntree (1975).

The salt marsh of Bolinas Lagoon is typical of northern California salt marshes in terms of species composition (Macdonald & Barbour 1974). *Spartina foliosa* often marks the lowest margins of the marsh, but in some areas *Spartina foliosa* is absent and stands of *Salicornia virginica* constitute the lower edge of the marsh. At mid-tidal heights, the marsh is occupied by what Macdonald and Barbour (1974) have described as a *Salicornia virginica* zone. *Salicornia virginica* is common in this region and in some areas forms monocultural stands. Several other species, including *Distichlis spicata*, *Frankenia grandifolia*, *Jaumea carnosa*, *Limonium californicum*, *Plantago maritima*, *Spergularia media*, and *Triglochin maritima*, also occur in this zone, both in single species patches and in intermixed assemblages.

Salicornia virginica is not found in the upper reaches of the marsh. There the other species found in the middle marsh are joined by species such as *Grindelia stricta* and *Atriplex patula* var. *hastata*. My research sites were all located in the mid-tidal height region of the marsh (+ 1.70 to 1.83 m MLLW). For the remainder of this paper, the species will be referred to by their generic names.

Yearly rainfall patterns

The California coast has a Mediterranean climate which is typified by a relatively warm wet winter (October–April) and is followed by a relatively cool summer with little or no rain (Ornduff 1974). The official rainfall year runs from July 1 to June 30 in California.

Rainfall data were provided by the Point Reyes Bird Observatory Terrestrial Biology Program. The data were collected at their Palomarin field station which is approximately 3 km from Bolinas Lagoon. It is readily apparent that the amount and pattern of annual rainfall during this study was highly variable (Table 1). In an average year, most of the rain falls between November and March.

Figure 1 shows the amount of monthly rainfall that occurred at the field station during the course of this study. The percentile score for the rainfall

Table 1. Yearly rainfall totals for each year of the study and average yearly rainfall for the 24 year period from July 1965 to June 1990 as measured at the Point Reyes Bird Observatory Terrestrial Biology Program's field station at Palomarin, California. Also shown are percentiles for total rainfall calculated from the 24 years of data. Low percentiles occur for years with low rainfall, high percentiles for years with high rainfall.

Year	Total rainfall (cm)	Percentile
1986–87	64.15	28
1987–88	68.38	36
1988–89	85.60	56
1989–90	52.88	12
Annual mean for 24 year period	86.51	

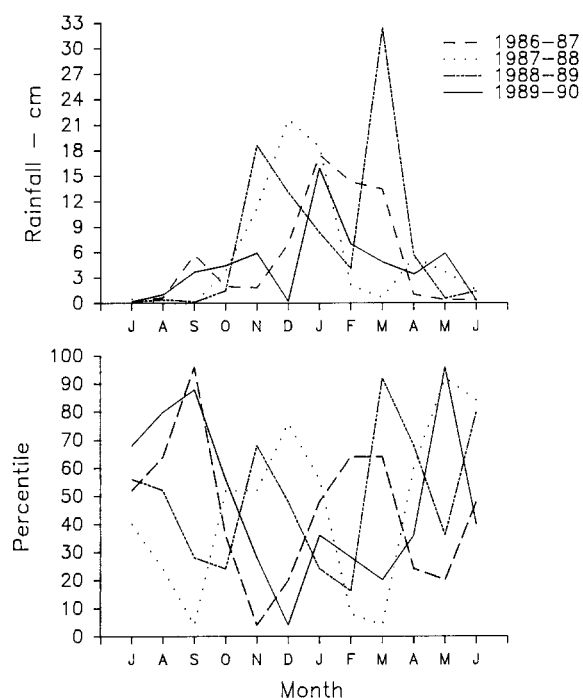


Fig. 1. A. Monthly rainfall totals in cm for each year of the study. B. Percentile scores for each month calculated from 24 years of rainfall data for the period of July 1965 to June 1990. High percentiles occur for months with relatively high rainfall, low percentiles for months with relatively low rainfall.

each month is also shown. Percentiles were calculated based on 24 years of rainfall collected from July 1965 to June 1990. Median rainfall has a percentile score of 50. In 1986–87, September had relatively high rainfall, but November and December had relatively low rainfall. In 1987–88 October, November, December and January rainfall was above the median, while February and March were well below the median. 1988–89 had total rainfall (85.60 cm) that was close to the 24 year average total of 86.51 cm. November and December had nearly median rainfall, January and February were low, followed by the second wettest March on record. Over 30 cm of rain fell in March 1989 alone. In 1989–90 rainfall was above the median for the late summer and early fall months, the winter months were below the median, and followed by the wettest May on record.

Sampling

In July 1987, four 10 × 10 m plots were established in the mid-tidal height marsh on Kent Island. The plots were all located on flat, well drained sites that appeared not to have been disturbed in the recent past. The plots were situated in areas that at the start of the study had fairly similar soil salinities (43 to 50 ppt). Soil salinities were measured at the start of the study to insure that sites were not too different in physical characteristics. Soil salinity was measured by squeezing out a drop of interstitial water from a soil sample and determining the salinity with a hand held refractometer. Soil samples were taken from a depth of 5–7 cm. Two of the plots were located in areas in which the plant assemblage was dominated by *Salicornia*; other species were relatively uncommon. The other two plots were located in areas in a more diverse plant assemblage in which several species, including *Salicornia*, were common. At the start of sampling, these two assemblage types were found to differ significantly in relative abundance of species when tested with a two factor nested MANOVA, in which the plot factor was nested within the assemblage type factor (Tables 2, 3). Plots nested within an assemblage type were not significantly different from each other, although some species differed between plots.

Table 2. Total percent cover for each of the permanent plots at the time of the initial sampling in July 1987. Plots 1 and 3 are the *Salicornia* dominated plots and plots 2 and 4 are the mixed species plots.

Species	Plot number			
	1	3	2	4
<i>Salicornia virginica</i>	189.7	215.6	107.7	107.8
<i>Distichlis spicata</i>	0	0	21.7	8.6
<i>Frankenia grandifolia</i>	31.4	3.6	0	26.7
<i>Jaumea carnosa</i>	12.2	28.8	45.0	12.8
<i>Limonium californicum</i>	0	0.6	2.8	7.5
<i>Spartina foliosa</i>	0.8	0.3	0	0
<i>Spergularia media</i>	6.9	0	0	0.8
<i>Triglochin maritima</i>	3.1	5.3	17.8	23.9
Bare space	5.6	6.1	12.1	13.1

Table 3. The results of a two-way MANOVA with one nested factor, of assemblage composition for the four plots in July 1987 at the start of the monitoring. The independent factors are Type, the assemblage type in which the plot was located (either *Salicornia* dominated or mixed species assemblage) and Plot, the actual plot sampled. Plot is nested within assemblage type. Dependent variables are the species in the assemblage and bare space. The table shows the results of the univariate F tests for each species and the overall multivariate test for each independent factor.

F test	Type			Plot {Type}		
	DF	F	p	DF	F	p
<i>Salicornia</i>	1,32	19.480	0.000	2,32	0.361	0.700
<i>Distichlis</i>	1,32	18.015	0.000	2,32	3.350	0.048
<i>Frankenia</i>	1,32	0.156	0.695	2,32	3.335	0.048
<i>Jaumea</i>	1,32	0.774	0.386	2,32	3.667	0.037
<i>Limonium</i>	1,32	6.171	0.018	2,32	1.476	0.244
<i>Spartina</i>	1,32	2.909	0.098	2,32	0.727	0.491
<i>Spergularia</i>	1,32	1.496	0.230	2,32	1.960	0.157
<i>Triglochin</i>	1,32	20.077	0.000	2,32	0.764	0.474
Bare space	1,32	13.198	0.001	2,32	0.087	0.917
Multivariate Wilk's Lamba	9,24	6.696	0.000	18,48	1.523	0.123

The plots were sampled four times a year at three month intervals, starting in July 1987 and continuing until the final samples were taken in August 1990. The plots were sampled using a stratified random sampling design (Greig-Smith 1983) in which the plots were divided into 9 equally sized strata. In each stratum the position from which the samples were actually taken was chosen using a random numbers table. Sampling was done using a square point frame of 0.25 m². The frame had 100 holes drilled through it in a grid pattern, with the holes separated by 5 cm from each other. 40 holes were randomly chosen and the point sampling was done by dropping a metal pin (1.8 mm diameter) through those 40 holes. Every contact of the pin by a plant was recorded, with the contacts ordered from highest to lowest so that the canopy layer could be identified for later analysis. If no plant contacted the pin, that pin was recorded as having contacted bare ground. The number of pin contacts for each species in each sample were converted to percent cover values for data analysis. Data was analyzed in terms of percent total cover, in which all pin contacts were utilized and cover values can sum to well over 100%. Total percent cover values were used because some species were occasion-

ally found only in the understorey and thus would be missed if canopy values were used and because I felt that total cover was a more biologically relevant measure of plant abundance. In any case, the canopy exhibited similar dynamics to total cover (Allison 1991).

Analysis

Species richness, Shannon-Wiener diversity and dominance values for all 9 point frame quadrats sampled in each plot at each sampling date were calculated from total percent cover values. For the statistical analyses, I compared data from different years within a season to eliminate from the analyses any variation due purely to seasonal differences. I chose to compare data collected during the summer, because I had more years of data collected during the summer than for the other seasons. Seasonal patterns are described, but were not investigated with statistical tests.

Homogeneity of the variances for species richness, Shannon-Wiener diversity and dominance values were tested using Cochran's C test (Winer 1971; pp. 208). All of these data sets were found to have homogeneous variances. Three factor

ANOVA's, with sampling date, assemblage type and plot (nested within assemblage type) as the independent variables, were done to examine whether or not the plots changed over time. Because the point frame was randomly placed to sample different areas of each strata in each plot at each sampling date, and because pin positions were randomly assigned for each sampling date, data collected at each sampling date represented an independent sample.

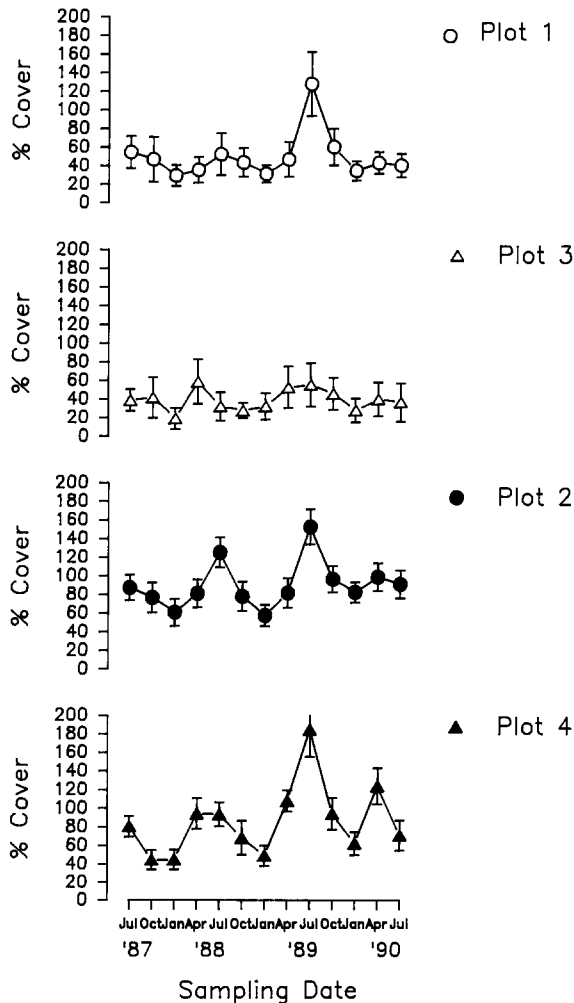


Fig. 2. Total percent cover summed for all species shown for all sampling dates from July 1987 to August 1990. Plots 1 and 3 are from the *Salicornia* dominated assemblage type and plots 2 and 4 are from the mixed species assemblage type. Error bars represent one standard error.

Results

Temporal changes in plant cover

Total plant cover (summed for all species) was generally highest in the spring and summer of each year (Fig. 2). In three of the plots total plant cover was highest in the summer of 1989, the year with the wettest spring. However plot 3, a *Salicornia* dominated plot, slightly declined in maximum total plant cover throughout the study.

Total plant cover summed for all species exclusive of *Salicornia*, shows this collection of less

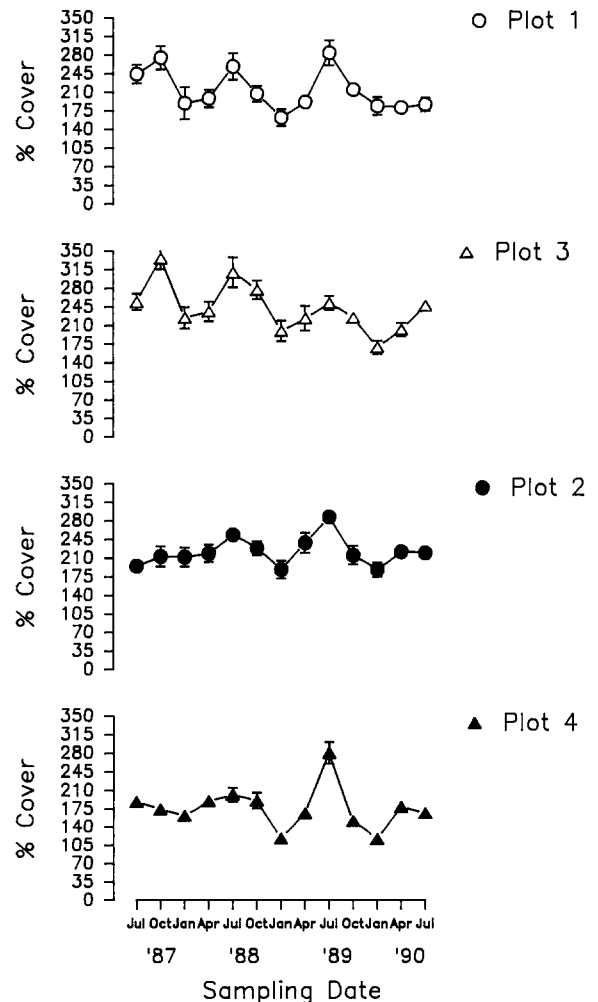


Fig. 3. Total percent cover summed for all species exclusive of *Salicornia* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

common species had their highest cover in the summer (Fig. 3). Following the heavy rainfall in the spring of 1989, the wettest year of the study, 3 of the plots had their highest cover of the less dominant species. Plot 3, the remaining plot, did not show any obvious pattern of year-to-year variation.

Because *Salicornia* has a somewhat longer growing season than the other species (Purer 1942; Cameron 1972; Macdonald 1977), its maximum total cover often occurred in the autumn (Fig. 4). Three of the plots (1 and 3 which are *Salicornia* dominated and plot 4 which is a mixed

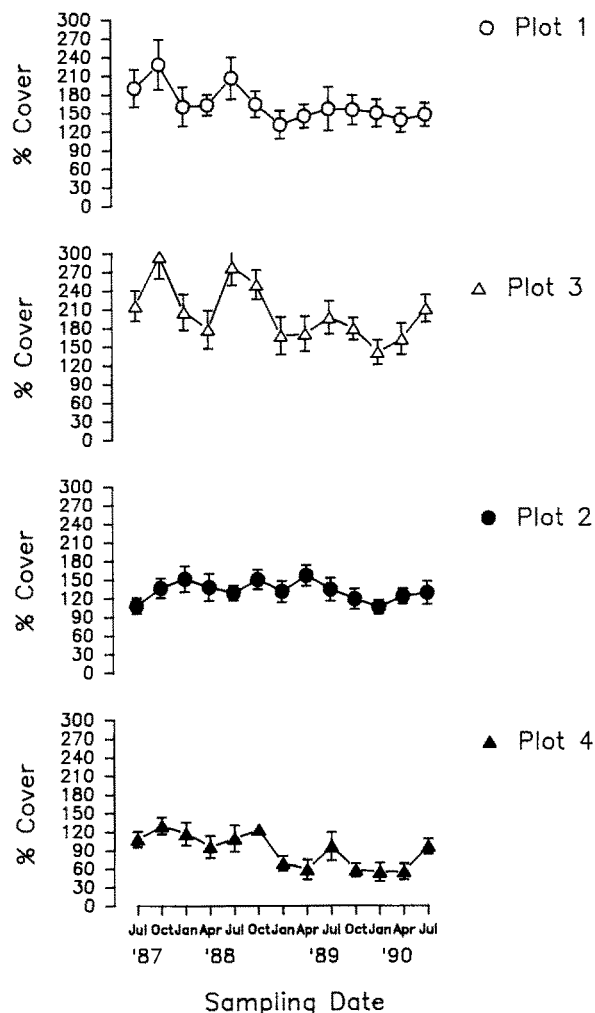


Fig. 4. Total percent cover of *Salicornia* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

species plot) showed a slight decline in the total cover of *Salicornia* during the time of this study, with the lowest amount of cover occurring in 1989 followed by a slight rise in total cover during 1990. Plot 2 showed no clear yearly trend.

In the mixed species assemblage the greatest total cover for *Distichlis* usually occurred in the spring or summer (Fig. 5). There was a gradual increase in the maximum yearly total cover values for *Distichlis* throughout the study. In plots 2 and 4, the maximum amount of total cover occurred during the summer of 1989.

Patterns of changes in total cover were highly variable for *Frankenia* within and among years, although the highest total cover generally occurred in the spring and summer (Fig. 6).

Jaumea, which was common in all plots, had its highest total cover in the spring and summer (Fig. 7). There was no obvious year-to-year trend in maximum total cover, but *Jaumea* had its maximum total cover for all plots in the spring or summer of 1989.

Spergularia was patchy in its distribution across the marsh (S. Allison pers. obsv.). *Spergularia* occurred in plot 1, a *Salicornia* dominated plot. It had a distinct seasonal growth pattern (Fig. 8). The greatest amount of cover occupied by *Sper-*

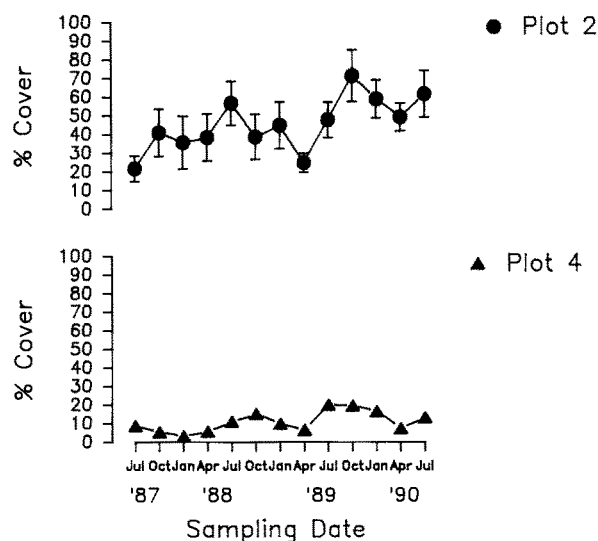


Fig. 5. Total percent cover of *Distichlis* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

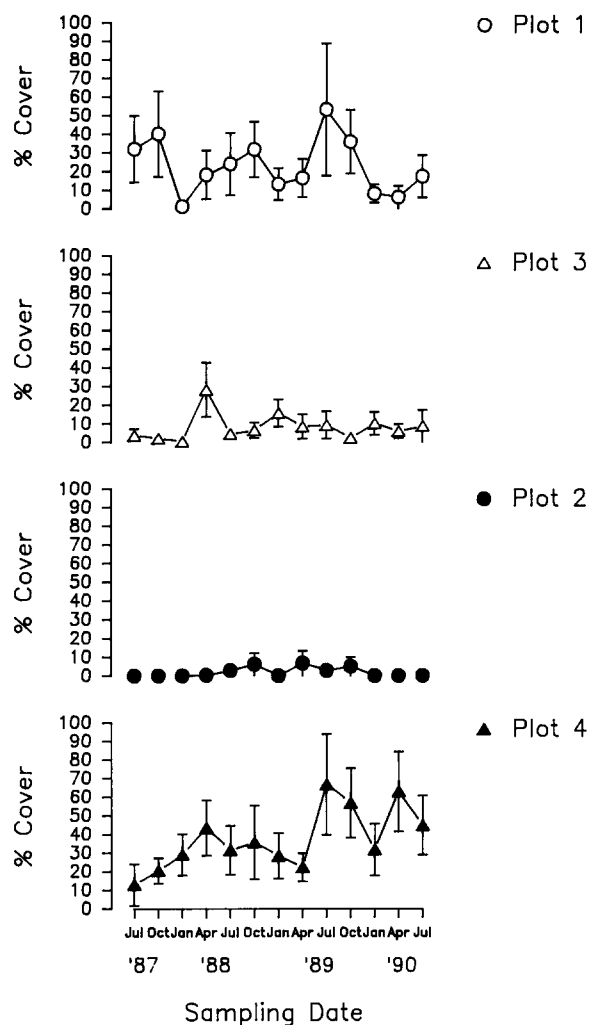


Fig. 6. Total percent cover of *Frankenia* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

gularia occurred in the spring and summer of 1989.

Triglochin had the most distinct seasonal growth pattern (Fig. 9). It was only commonly found in the mixed species assemblage. The maximum total cover values for *Triglochin* occurred in the summer of 1989.

Richness, diversity and dominance

Seasonal variation

Species richness and diversity were lowest in the winter and fall and highest in the spring and sum-

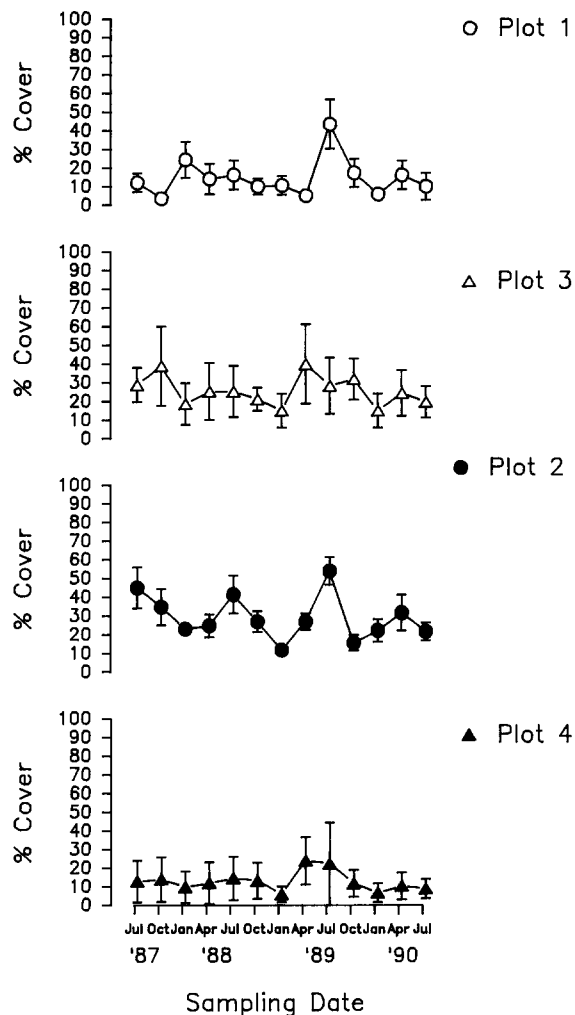


Fig. 7. Total percent cover of *Jaumea* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

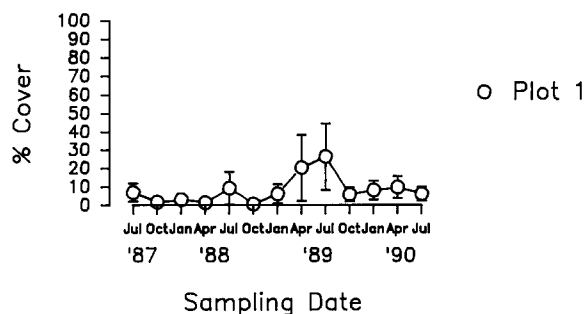


Fig. 8. Total percent of *Spergularia* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

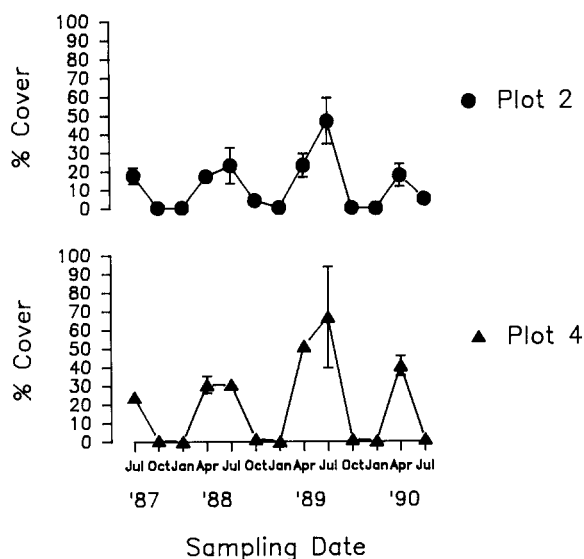


Fig. 9. Total percent cover of *Triglochin* for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

mer (Fig. 10 and 11). In contrast, dominance was lowest in the spring and summer and highest in the fall and winter (Fig. 12).

Interannual variation

Summer values of species richness, diversity and dominance all differed significantly between as-

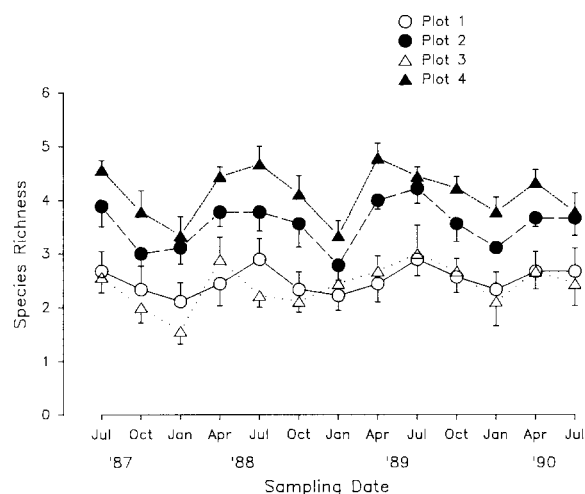


Fig. 10. Species richness for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

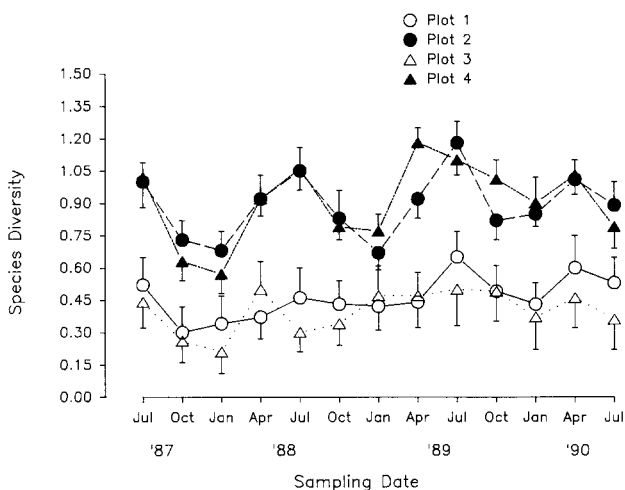


Fig. 11. Shannon-Wiener diversity for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

semblage types (Tables 4, 5, 6). There was also a strong tendency for dominance ($p = 0.059$) and diversity ($p = 0.068$) to differ among years suggesting that the unusually low dominance and high diversity of 1989 was probably not due to chance. Plots nested within an assemblage differed in summer dominance but not in species richness or diversity.

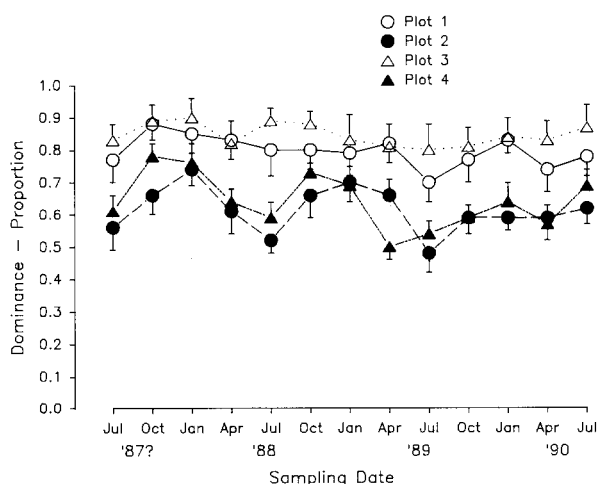


Fig. 12. Dominance for all sampling dates from July 1987 to August 1990. See Figure 2 for key to symbols.

Table 4. Results of three factor analysis of variance in species richness for plots sampled during the summers of 1987–90.

Source of variation	DF	SS	MS	F	p
Year	3	4.521	1.507	1.380	0.252
Asbg. type	1	76.562	76.562	70.111	0.000
Plot {Asbg. type}	2	4.903	2.451	2.245	0.110
Year × Asbg. type	3	1.410	0.470	0.430	0.732
Year × Plot {Asbg. type}	6	3.264	0.544	0.498	0.809
Error	128	139.78	1.092		

Table 5. Results of three factor analysis in Shannon-Wiener diversity for plots sampled during the summers of 1987–90.

Source of variation	DF	SS	MS	F	p
Year	3	0.861	0.287	2.427	0.068
Asbg. type	1	10.628	10.628	89.879	0.000
Plot {Asbg. type}	2	0.379	0.190	1.603	0.205
Year × Asbg. type	3	0.365	0.122	1.030	0.382
Year × Plot {Asbg. type}	6	0.078	0.013	0.109	0.995
Error	128	15.135	0.118		

Table 6. Results of three factor analysis of variance in dominance for plots sampled during the summers of 1987–90.

Source of variation	DF	SS	MS	F	p
Year	3	0.234	0.078	2.539	0.059
Asbg. type	1	1.904	1.904	61.937	0.000
Plot {Asbg. type}	2	0.202	0.101	3.290	0.040
Year × Asbg. type	3	0.077	0.026	0.837	0.476
Year × Plot {Asbg. type}	6	0.006	0.001	0.031	1.000
Error	128	3.936	0.031		

Discussion

Seasonal patterns

The clear seasonal patterns (Figs. 2–9) of active growth in spring and summer and dormancy (*sensu* Harper 1977; pp. 62) in the fall and winter have been observed before in California salt marshes and are probably typical of salt marshes on the west coast of North America (Purer 1942; Cameron 1972; Mahall & Park 1976a; Macdonald 1977; Zedler 1982). Most of the plants do not initiate their growth until after the winter rainy season is almost over. During the summer the only water usually available to the plants is salt water which arrives through tidal action. Many of

the species are better able to expand via vegetative growth after freshwater has flushed salt from the soil. High soil salinity may delay the onset of growth until after most of the rains have occurred (Zedler & Beare 1986; Zedler *et al.* 1986).

The changes in species growth lead to seasonal patterns in species richness, dominance and species diversity that are consistent from year to year and from place to place within the marsh. These patterns are consistent despite variation in abundance and the amount of cover occupied.

Patterns in species richness are primarily due to seasonal differences in growth of the plants. *Triglochin* dies back to the soil surface during the fall and winter and regrows from persistent rhizomes in the spring and summer. *Spergularia* loses

much of its aboveground biomass in the fall and winter as well. It grows back from persistent rootstocks in early spring and has considerable aboveground biomass throughout the summer. All of the species are dormant in the fall and winter and lose dead plant tissue which is usually carried away by high tides. When species occurring at low abundance at a particular site die back, leaving only a small amount of aboveground biomass, they are not as apparent to point sampling in the fall and winter. This will result in lower estimates of species richness at those times of the year.

Seasonal changes in dominance are largely due to changes in the percent cover of *Salicornia* relative to the percent cover of the other species. *Salicornia* was the dominant species in terms of cover in almost all of the quadrats sampled (Fig. 4). The other species all experience a greater increase in aboveground biomass during the spring and summer growing season than does *Salicornia*. *Salicornia* has persistent woody stems and rootstocks and does not lose as much aboveground biomass in the fall and winter as do the other species. This causes a relative shift in the degree of dominance of *Salicornia*, such that it is most dominant in the fall and winter. All of the plots were found to have their lowest dominance values during the spring and summer of 1989.

Patterns in diversity are due to a combination of changes in the number of species present aboveground and to changes in the relative degree of dominance by *Salicornia*. The highest diversity values for all plots occurred during the spring and summer of 1989. In plot 3, a *Salicornia* dominated plot, equivalently high diversity values occurred in April 1988 and July 1989.

Year to year variation

A comparison of the dominance and diversity values obtained for the summer sampling dates revealed distinct variation between years. Diversity was highest in the summer of 1989 and conversely, dominance was lowest at that time. Richness was fairly consistent from year to year within

a particular season. The variation in diversity and dominance seems to be related to variation in winter and spring rainfall. Rainfall totals for the years 1986–87, 1987–88 and 1989–90 were below average. All of those years had periods of 2 to 3 months in the winter when little rain fell. 1988–89, the water year with the highest diversity and lowest dominance values, had average rainfall in the fall and much greater than average rainfall in March. Diversity increased and dominance decreased during 1989 because most of the plant species became more abundant relative to *Salicornia*.

Summer rainfall is unusual along the California coast even during normal years. As a consequence, the salinity of salt marsh soil typically becomes elevated during the summer as increased evaporation removes water from the soil surface leaving behind the salts (Purer 1942, Mahall & Park 1976b; Macdonald 1977; Zedler 1982; Zedler & Beare 1986). During the summer the only source of water is seawater that is brought in by the tides. The middle and upper marsh are inundated by tides only a few days each month and evaporation rates can be very high in the summer, resulting in salts being carried up from deep in the soil as water is drawn up and evaporated away. Rainfall will provide an input of freshwater and flush out the salt concentrated at the soil surface during a winter of average rainfall although annual variation in rainfall pattern prevents most years from matching an average pattern (Michaelsen *et al.* 1987).

Yearly variations in the relative abundances of species were consistent with known differences in their physiology. *Salicornia*, which grew relatively better during years with extended periods of low rainfall, is better able to tolerate hypersaline soil conditions than are the other species (Barbour & Davis 1970; Mahall & Park 1976b; Percy & Ustin 1984; Zedler & Beare 1986). Thus *Salicornia* will tend to grow faster, accumulate more biomass and occupy more cover than will the other species during times of high soil salinity (Zedler & Beare 1986). The other species (Figs. 5–9) grew relatively better during the one year (1988–89) when heavy rainfall occurred during the early part

of the growing season. During the wet year, *Salicornia* did not grow to occupy as much cover as during the other years. This is especially evident in plots from the *Salicornia* dominated assemblage type (Fig. 4).

Almost all halophytes will grow faster and accumulate more biomass when provided freshwater while growing (Pigott 1969; Barbour & Davis 1970; Phleger 1971; Mahall & Park 1976b; Barbour 1978; Storey & Wyn Jones 1979; Cooper 1982; Okusanya & Ungar 1984; Percy & Ustin 1984; Snow & Vince 1984). *Distichlis*, *Frankenia*, *Jaumea*, *Salicornia* and *Triglochin* all have been shown to grow better when freshwater is available (Pigott 1969; Barbour & Davis 1970; Barbour 1978; Cooper 1982; Snow & Vince 1984). In a direct comparison of growth in soil watered with freshwater in a greenhouse, *Frankenia* and *Distichlis* both experienced more growth than did *Salicornia*, though *Salicornia* had more growth than did *Jaumea* (Barbour & Davis 1970). In another greenhouse study, *Triglochin* grew better in freshwater than did *Salicornia europaea*. *S. europaea* grew best in high salinity soils (Cooper 1982). The less dominant species are probably more dependent on having freshwater available to them in order to initiate growth. They experience elevated growth relative to *Salicornia* when more freshwater is available. It is likely that the cover values obtained in the summer of 1988–89 are typical of the relative amounts of cover that exist during years when periods of heavy rainfall flush large amounts of salt from the soil. It is possible that during extended periods of much greater than average rainfall, the marsh would come to be dominated by species that now appear to be uncommon relative to *Salicornia*. Such changes in species composition due to freshwater flooding have been documented for salt marshes in southern California (Zedler 1983; Zedler *et al.* 1986).

Influence of rainfall variability

Rainfall records for the California coast over the past 400 years, as inferred from tree-rings, indi-

cate that the period from 1920–65 was a time of unusual stability in rainfall (Michaelsen *et al.* 1987). Since that time there has been increasing variability in the amount of rainfall with extremely dry (1976–77) and extremely wet (1982–83) years occurring recently.

Extreme meteorological events may exert lasting changes in the species composition of California salt marshes (Zedler & Onuf 1984). Such changes have been documented to occur after unusual freshwater flooding has occurred in southern California marshes (Zedler 1983; Zedler *et al.* 1986). Long term effects have been caused by these flooding events because some species were unable to tolerate the lengthy inundation and submergence and thus died. Once the species were removed from the marsh, they have been slow to reinvade either because of the loss of a viable seed bank for an annual or the inability of seedlings of halophytic perennials to compete with established plants, such as *Typha dominguensis*, which can tolerate high salinity soil and water but whose seeds require brackish water to germinate (Zedler 1983; Zedler *et al.* 1986). *Spartina* has been shown to suffer increased mortality during a severe one year long drought in southern California (Zedler *et al.* 1986). Changes in the relative amounts of freshwater and saltwater input into salt marshes have also been implicated in rapid changes in species composition in New England (Niering & Warren 1980; Clark & Patterson 1985) and in the Netherlands (Olf *et al.* 1988).

Salt marsh soil salinity can be influenced by a complex set of interactions between rainfall and tidal inundation. The amount of rainfall, timing of rainfall, level of tidal inundation, timing of the highest tides and proximity to freshwater streamflows are all important factors which determine soil salinity (Olf *et al.* 1988; De Leeuw *et al.* 1990; Gross *et al.* 1990). Soil salinity probably will not be lowered if rainfall occurs during times of high tides and when the soil is saturated with salt water. The soil salinity of areas of marsh that are far away from freshwater streamflows experience greater changes due to freshwater input from rainfall (Gross *et al.* 1990). Kent Island is situated in the center of Bolinas Lagoon and does not

have any freshwater streams flowing through it or near it. Soil salinity at Kent Island is dependent on the interaction between tidal inundation and rainfall. The large amounts of rain that fell in March 1989, primarily on the 2nd, 17th, 18th and 21st, occurred when the tidal cycle was approximately mid-way between the spring and neap tides (Tide Tables, U.S. Department of Commerce, 1989). The highest tides occurred early in the morning and were all +1.7 m or lower. The lowest tides occurred in the early afternoon and were all below +0.18 m. 6.1 cm of rain fell on the 2nd and 8.9 cm of rain fell on the 17th and 18th. The tidal range from +1.7–1.83 m MLLW is covered by a high tide at least once every 4 to 5 days during most of the year at Bolinas. However, during March and April there are periods of up to 18 days in which no tides higher than +1.7 m occur (Tide Tables, U.S. Department of Commerce). Rainfall which occurs in March and April will have an especially large effect on the plants. The marsh was exposed to the air during these rainfall events and there is a large probability that the rain flushed the soil of some of its salinity load.

Grime (1977) has defined stress as any agent or set of conditions which reduces growth and productivity. Middle marsh species other than *Salicornia* appear to be stressed by a lack of freshwater during most years. During 1988–89 when there was heavy rainfall at the start of the growing season, those species experienced an increase in cover relative to *Salicornia* due to a release from this stress. This release from stress due to a large input of freshwater is probably a relatively unusual event and the less abundant species at Bolinas probably depend upon these events to spread both vegetatively and by seedling establishment. Once long lived perennial species become established, they can survive high soil salinity until the next period of heavy rainfall occurs and allows them to expand again. Increases in species sensitive to high soil salinity have been observed to occur after unusually wet years in southern California (Zedler & Beare 1986).

Geographic generality

There are few long term studies of species composition in tidal marshes (Odum 1988). Several long term studies of species composition in grazed and mown marshes have been done in Europe (Andresen *et al.* 1990; De Leeuw *et al.* 1990; other references cited in those papers) but more studies of variability in species composition are needed from marshes that are in a natural state. This study has documented that species composition did not change significantly during a period of variable rainfall. Onuf (1987) also found that species composition was relatively constant throughout a 5 year period in Mugu Lagoon, southern California. He did notice a brief fluctuation in species composition that was related to a large input of freshwater and sediment by flooding. Long lasting changes in species composition following freshwater flooding (Zedler 1983; Zedler *et al.* 1986) and reductions of growth due to drought (Zedler & Nordby 1986; Zedler *et al.* 1986) have been observed in Tijuana Estuary. Stratigraphic studies of cores from marshes have documented that changes in species composition have historically been associated with changes in relative inputs of freshwater and salt water (Clark & Patterson 1985; Schwartz *et al.* 1986). These changes can be quick on a geological time scale, but it is hard to relate them to short term climatic fluctuations.

It is difficult to determine how general are the results of this study. Onuf (1987) and I have observed relatively constant species composition despite fluctuations in climatic conditions which might indicate that marshes are robust and can resist such changes. However the differences in climate between the two sites calls for caution in drawing general conclusions based on the two studies. Onuf (1987) was unwilling to make general conclusions about all of Mugu Lagoon because he had only studied the marsh in the eastern part of the lagoon and was not confident that the marsh in the western part would have similar dynamics. Josselyn (1983) felt that conditions around San Francisco Bay were so variable that there was no way to draw conclusions

about a typical San Francisco Bay salt marsh. There is considerable local variation among salt marshes in southern California (Purer 1942; Zedler 1982; Zedler 1986).

Variation among marshes is even greater over larger spatial scales. It is well known that marsh productivity and structure varies with latitude along the east coast of North America (Turner 1976). There is much variation in species composition among marshes on the Pacific coast of North America (Macdonald & Barbour 1974; Macdonald 1977). Variation between west and east coast marshes of North America (Zedler 1982; Josselyn 1983; Seliskar & Gallagher 1983; Onuf 1987; Odum 1988) is even larger and is mostly related to differences in rainfall patterns, tidal inundation and geographic extent of the marshes.

Given the state of global climatic change, it is important that we conduct numerous long term studies of species composition in undisturbed sites, not just in marshes, but in all habitats (Mooney 1991). Without such data we will not be able to detect change when it is occurring or to predict the direction of any change. Because marshes are variable on local and regional scales, it is important that we have many such studies recording similar data in order to be able to make generally applicable conclusions and predictions about their dynamics (Foster 1990).

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